



21 DAYS SQL CHALLENGE

DAY 20

BY AMRUHA AHMED

DAILY CHALLENGE

Create a trend analysis showing for each service and week: week number, patients_admitted, running total of patients admitted (cumulative), 3-week moving average of patient satisfaction (current week and 2 prior weeks), and the difference between current week admissions and the service average. Filter for weeks 10-20 only.

```
SELECT
week,
patients_admitted,
SUM(patients_admitted) OVER (PARTITION BY SERVICE ORDER BY week ASC) AS "Running Total of
Patients Admitted",
ROUND(AVG(patient_satisfaction) OVER (PARTITION BY SERVICE ORDER BY week ASC ROWS
BETWEEN 2 PRECEDING AND CURRENT ROW)) AS "Moving Average of Patient Satisfaction",
ABS(patients_admitted-ROUND(AVG(patients_admitted) OVER (PARTITION BY service ORDER BY week
ASC))) AS "Difference"
FROM services_weekly
WHERE week BETWEEN 10 AND 20;
```

AMRUHA AHMED

DAILY CHALLENGE

week	patients_admitted	Running Total of Patients Admitted	Moving Average of Patient Satisfaction	Difference
10	17	17	81	0
11	16	33	90	1
12	28	61	82	8
13	23	84	81	2
14	15	99	72	5
15	17	116	81	2
16	23	139	76	3
17	21	160	80	1
18	18	178	76	2
19	16	194	88	3

AMRUHA AHMED

PRACTICE QUESTION 1

Calculate running total of patients admitted by week for each service.

```
SELECT  
service,  
week,  
SUM(patients_admitted) OVER(PARTITION BY SERVICE ORDER BY week ASC) AS "Running  
Total"  
FROM services_weekly;
```

service	week	Running Total
emergency	1	32
emergency	2	60
emergency	3	92
emergency	4	124

AMRUHA AHMED

PRACTICE QUESTION 2

Find the moving average of patient satisfaction over 4-week periods.

```
SELECT  
service,  
week,  
AVG(patient_satisfaction) OVER (PARTITION BY service ORDER BY week ASC ROWS  
BETWEEN CURRENT ROW AND 3 FOLLOWING) AS "Moving Average"  
FROM services_weekly;
```

service	week	Moving Average
emergency	1	74.5000
emergency	2	81.0000
emergency	3	80.2500
emergency	4	78.2500

AMRUHA AHMED

PRACTICE QUESTION 3

Show cumulative patient refusals by week across all services.

```
SELECT
service,
week,
patients_refused,
SUM(patients_refused) OVER(ORDER BY week ASC) AS "Cumulative Patient Refusals"
FROM services_weekly;
```

service	week	patients_refused	Cumulative Patient Refusals
emergency	1	44	302
surgery	1	85	302
general_medicine	1	164	302
ICU	1	9	302
emergency	2	141	583

AMRUHA AHMED

The background features two large, flowing, and swirling liquid-like shapes on a solid black background. These shapes are highly reflective and iridescent, displaying a spectrum of colors including red, orange, yellow, green, blue, and purple. They appear to be made of a viscous, glossy material, possibly oil or paint, that has been manipulated to create these fluid, organic forms. The central text is positioned between these two decorative elements.

THANK YOU

for your time and attention

AMRUHA AHMED